UM Hackathon 2026

LintasNiaga

Trade with confidence • Secure against hidden market risk

by 404 Not Founders

Market Statistics

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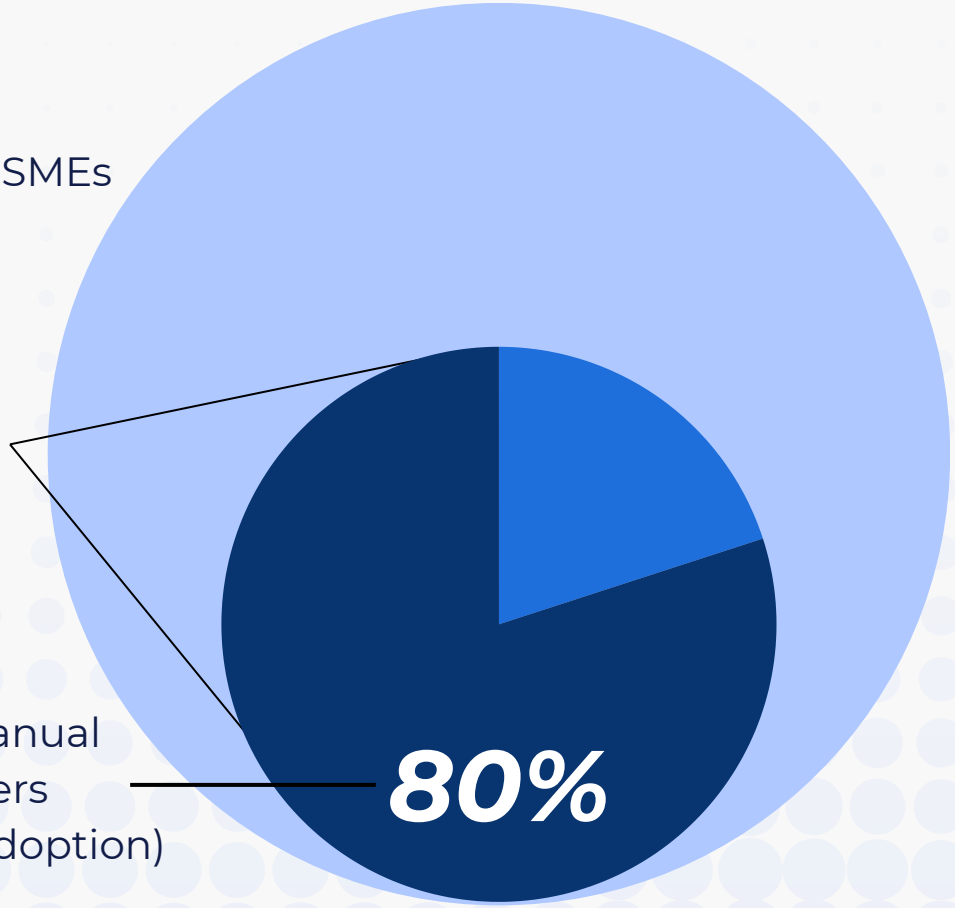
~85K Total Manufacturing SMEs

~1.5K Plastic Manufacturing SMEs

Target: ~1.2K Manual Decision Makers (Potential Digital Adoption)



RM 18.5B

Malaysia Plastics Industry Market Size

Source: Mordor Intelligence – Malaysia Plastics Market Analysis (2026–2031).

RM 4.2B

PP Resin Imports

Source: OEC - Propylene Polymers in Malaysia (2025/2026 Data).

RM 336M

Avoidable Margin Loss (8%)

Source: CIPS (Chartered Institute of Procurement & Supply) – Procurement ROI Benchmarks.

Source: MPMA Industry Census 2025/2026; Department of Statistics Malaysia (DOSM) MSME Performance Report.; MDEC – Malaysia Digital Economy Blueprint (MyDIGITAL) 2025 Targets.

Problem Statement

The Hidden Costs Behind Every Quote

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Currency Volatility

MYR/USD can shift 3–5% between quote receipt and payment, silently eroding margins.

Commodity-Linked Shocks

Oil and petrochemical movements affect both material cost and shipping cost, requiring real-time market awareness.

Supplier Reliability Risk

A cheaper quote becomes expensive if the supplier delivers late or inconsistently.

MOQ Lock-Up

Low unit price hides capital waste when minimum order quantities exceed real needs.

Timing Uncertainty

The right supplier may differ depending on urgency or upcoming market events.

Logistics Disruption

Holidays, port delays, route disruptions, or severe weather can quietly break an apparently good deal.

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Your Strategic Procurement Copilot,
transforming market volatility into a competitive advantage
through data-driven precision and secure, traceable trade decisions.

Core Features

Finding The Best Value for SMEs

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01 **EXTRACT**
Intelligent PDF parsing converts messy supplier quotes into structured data using computer vision

02 **COMPARE**
Comparison across price, terms, reliability and delivery timelines with normalized scoring algorithms

03 **ANALYSE**
AI analyst interprets oil price fluctuations and macro trends in real-time

04 **HEDGE**
Recommend optimal hedge ratios based on FX volatility, urgency and buyer preference

Tech Stack

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FRONTEND

- Next.js 16
- React 19
- TypeScript
- Tailwind CSS
- Base UI
- React Query
- Recharts
- React PDF

Decision workspace, upload flow, fan chart, hedge slider, PDF output

BACKEND

- Python 3.12+
- FastAPI
- Uvicorn
- Pydantic v2
- PyMuPDF
- httpx
- tenacity

Workflow owner, quote extraction, validation, APIs, SSE

DECISION ENGINE

- NumPy
- Pandas
- yfinance
- FX + Brent Monte Carlo
- p10 / p50 / p90 risk bands
- deterministic ranking

Landed cost, supplier ranking, risk simulation, hedge replay

AI & TRACEABILITY

- GLM model endpoint
- LangChain messages
- thin LangGraph node
- Langfuse

Strict JSON reasoning, AI explanation, traceable recommendation

DATA LAYER

Web Scraping Pipeline

- SunSirs PP resin benchmark
- Trafilatura + BeautifulSoup parser
- Raw HTML/text evidence
- Normalized resin snapshot

Market & Reference Data

- FX, oil, weather, macro, news, holidays
- Freight / tariff / supplier references

System Architecture

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1. User / SME

Procurement Team

- Upload supplier quote PDFs
- Enter quantity and urgency
- Review extracted quote fields

2. Next.js Frontend

Thin UI Layer

- Upload page
- Review & repair page
- Analysis progress page
- Results dashboard
- Shows supplier ranking, fan chart, hedge slider, AI explanation, bank-instruction PDF

3. FastAPI Backend Decision Engine

Core Brain – Owns Business Logic

- Quote extraction & validation
- Reference & snapshot loader
- Landed-cost calculator
- Supplier ranking engine
- Monte Carlo risk simulation
- Hedge replay
- Recommendation assembler

Deterministic math runs before AI reasoning.

5. GLM + Langfuse AI Layer

Bounded & Traceable AI

- GLM / AI reasoning
- Thin LangGraph orchestration
- Langfuse traceability

AI explains and reasons over structured backend results. It does not replace the cost engine.

Final Decision Output

- ✓ Recommended supplier
- ✓ Lock now or wait
- ✓ Suggested hedge %
- ✓ p10 / p50 / p90 landed-cost risk bands
- ✓ Why this supplier
- ✓ Why not the others
- ✓ Traceable AI reasoning link

4. Data & Market Context Layer

Supports Backend Analysis

- Uploaded quote PDFs
- Reference data: freight, tariff, supplier reliability
- FX and Brent oil snapshots
- Weather and holiday risk
- Macro and news snapshots
- SunSirs PP resin benchmark

External data is normalized into snapshots before analysis.

Product Showcase

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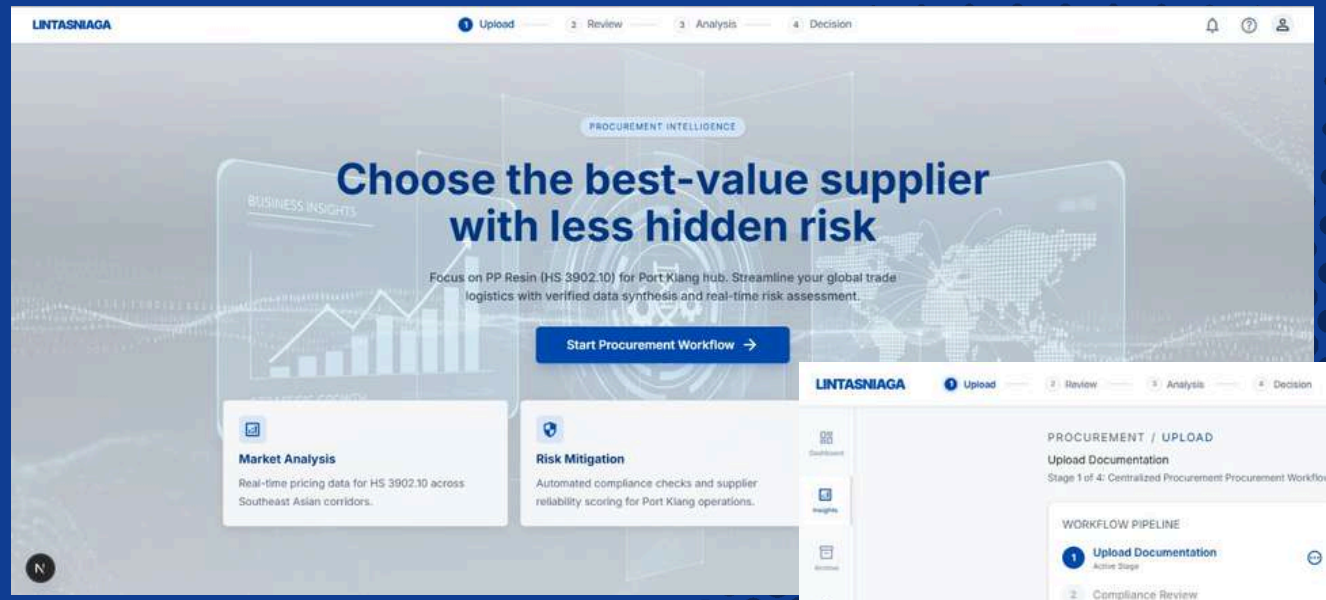
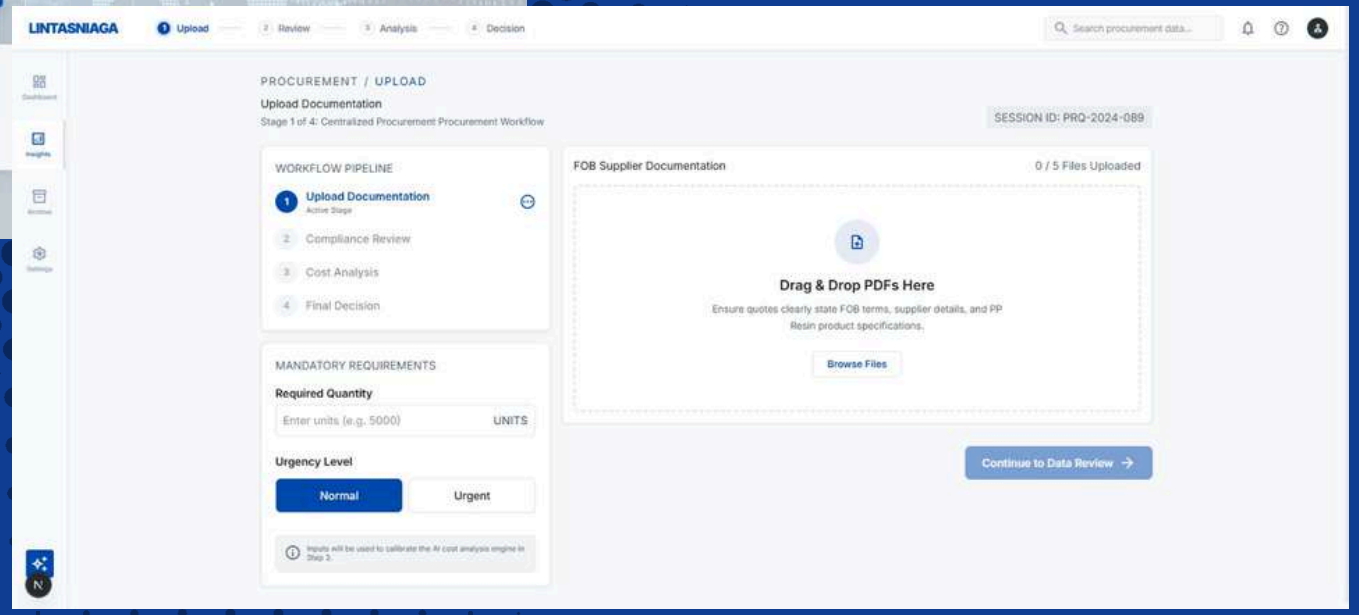


LintasNiaga Home Page

01 Upload Phase

Upload Your Quotation PDF, Confirm Required Quantity



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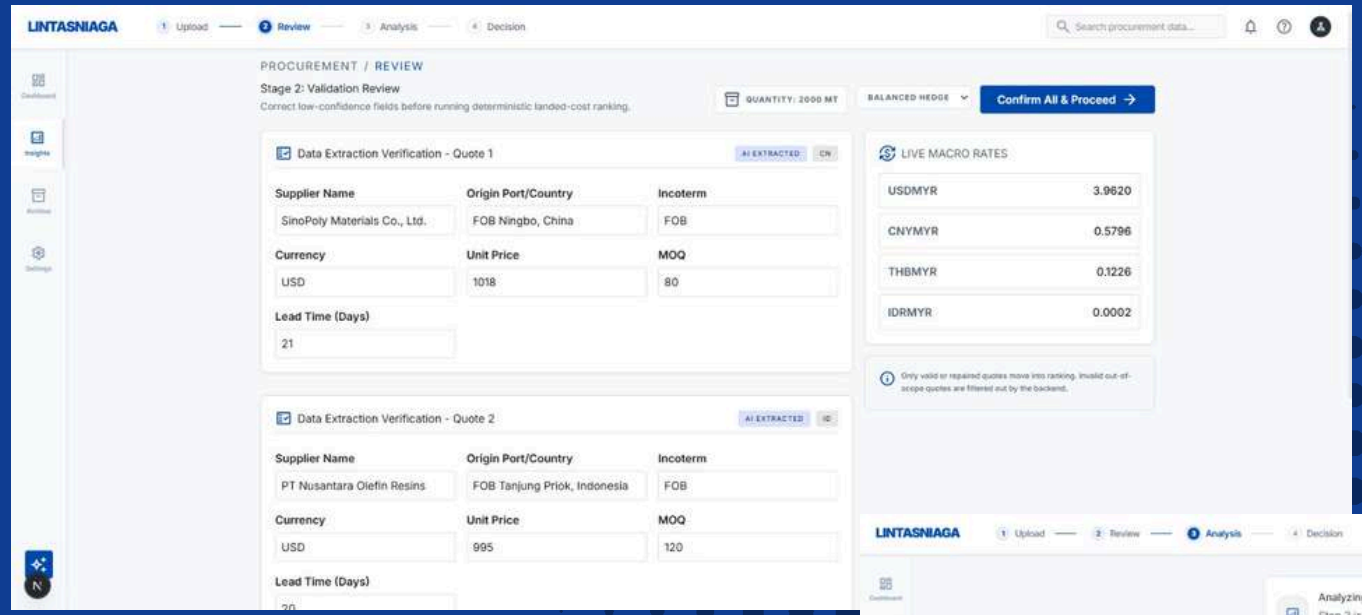
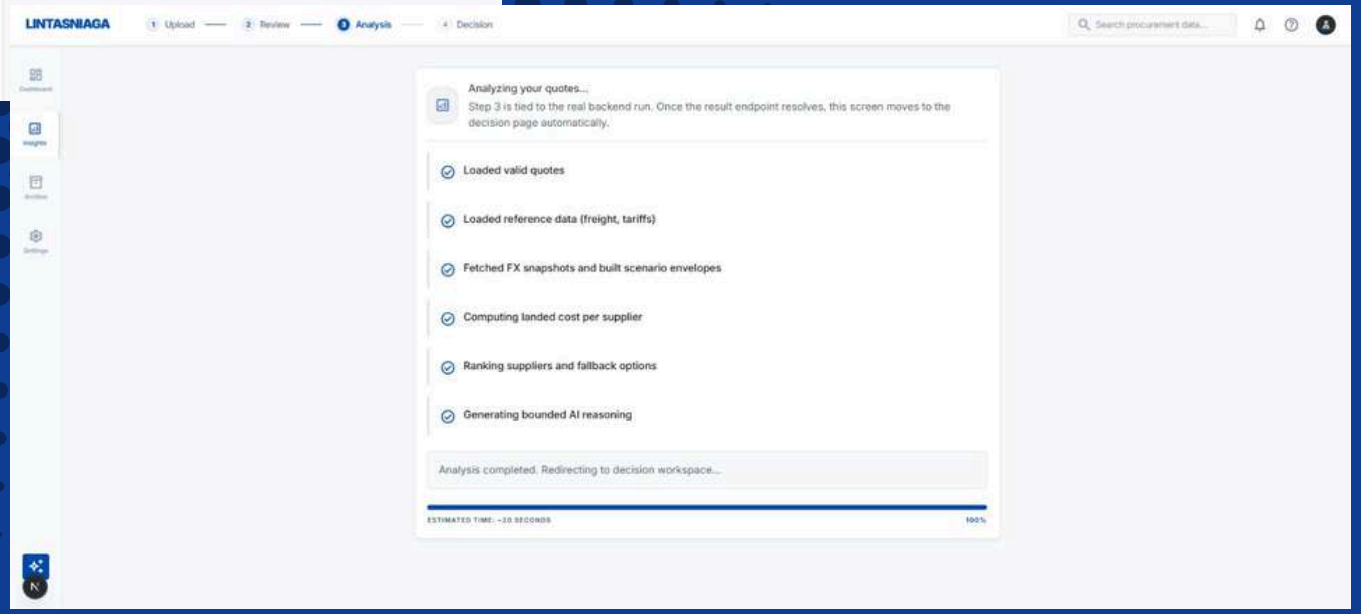


02 Review Phase

Data Validation Review by User

03 Analysis Phase

Model Analyzing



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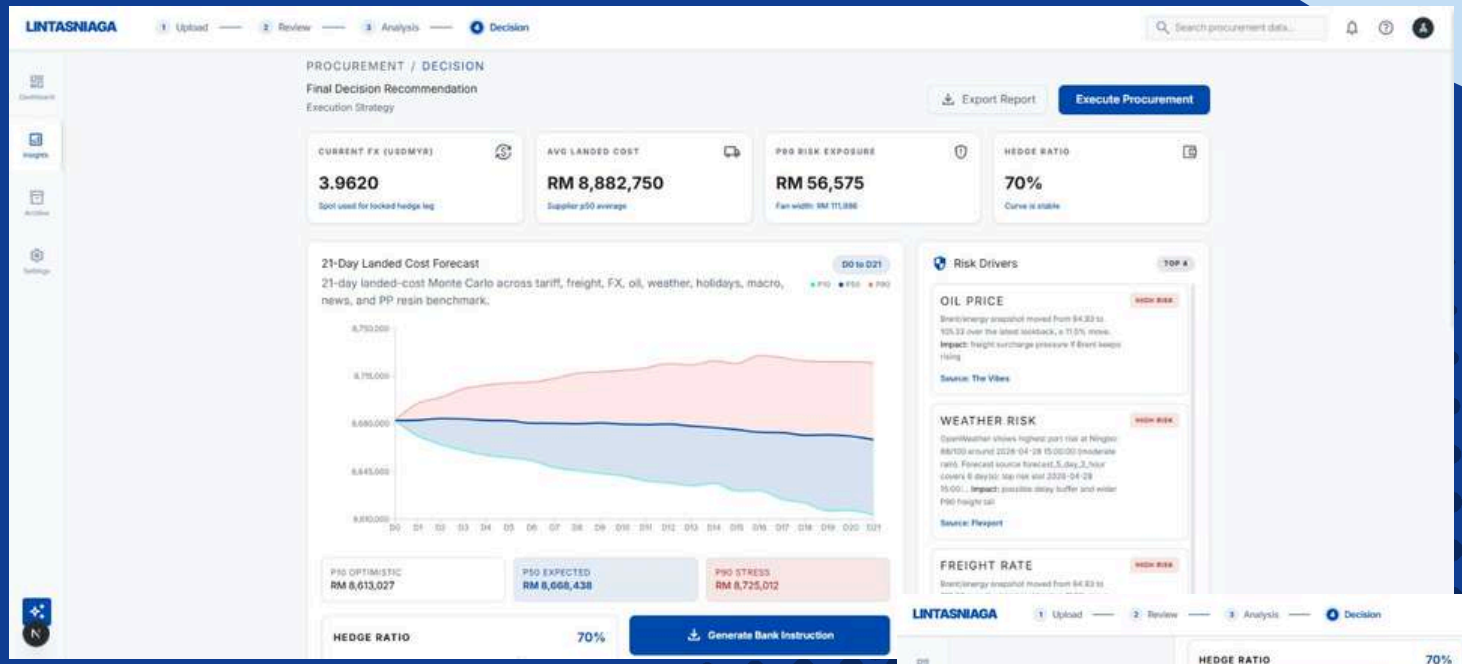
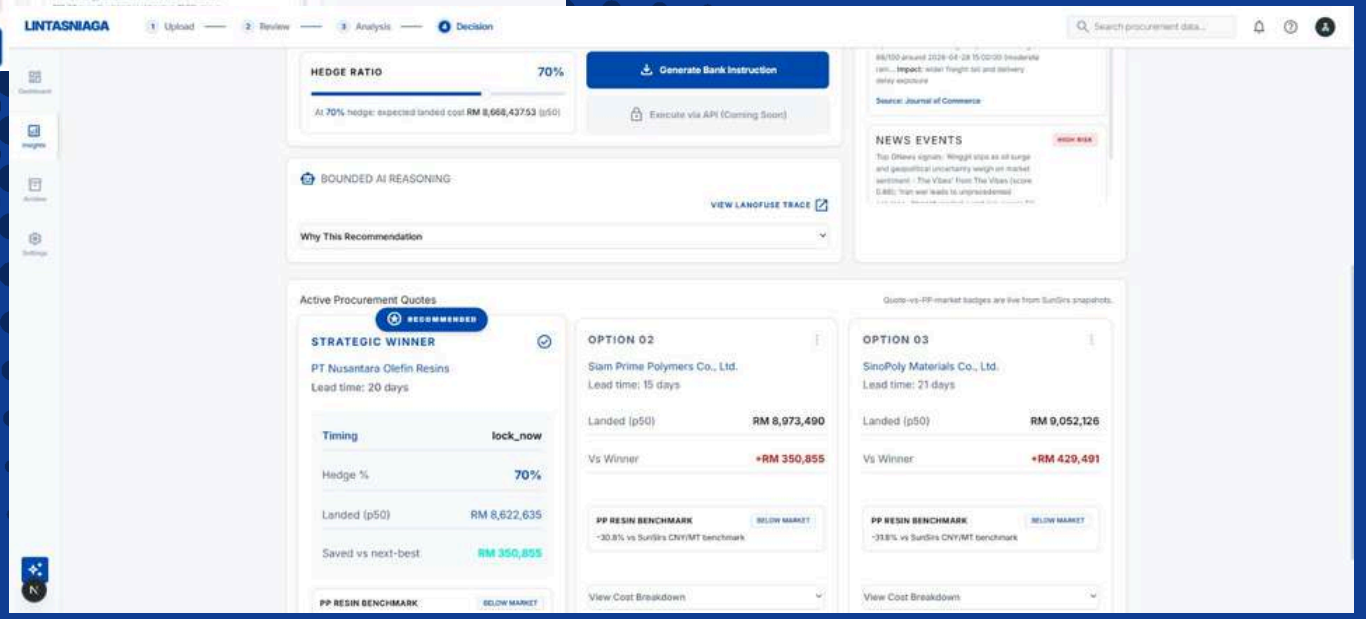


04 Decision Phase

Best Supplier Recommendation & Risk Drivers Reasoning

04 Decision Phase

Price Forecasting & Hedging Percentage Preferrence



Competitive Analysis

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	Spreadsheet & Email	Generic ERPs	Generic AI Assistants	LintasNiaga
SME-ready pricing	✓	✗	✓	✓
True MYR landed cost	✗	Partial	✗	✓
FX risk simulation	✗	✗	✗	✓
Supplier reliability scoring	✗	✓	✗	✓
Plain-language AI reasoning	✗	✗	Partial	✓
Hedge recommendation	✗	✗	✗	✓
PDF quote ingestion	✗	Partial	Partial	✓
Malaysia PP Resin–specific	✗	✗	✗	✓

Key Innovations

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Math-Verified Accuracy

Core math (landed costs, risk penalties and supplier rankings) are computed by a precise engine before AI reviewing.

Expert Industry Analysis

Our tool is specialized in specific material types and trade corridors in the Malaysian Plastic Industry context.

Comprehensive Risk Forecasting

The system models best and worse-case outcomes for a realistic view of potential risks (currency volatility and shipping delays).

Real-time System Reliability

Market data grounding by live-refreshed market snapshots, ensuring procurement decisions are based on verified data.

Transparent Decision Logic

Recommendations are accompanied by three plain-language reasons, bound to logical rules and traceable.

The Role of GLM

Essential Decision-Making Model

- Extract structured fields from supplier quote PDFs (vision mode).
- Reason over deterministic outputs: cost, risk, ranking.
- Generate plain-language explanations for recommendations.
- Apply bounded override logic between top-2 suppliers only.
- Convert news headlines into a structured event context.
- Extract PP resin benchmark prices from the curated web text.

Business Value

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For SME Procurement Teams

Replace manual decisions with structured, explainable analysis. Reduce costly errors in every import cycle.

For the Ecosystem Partner

LintasNiaga integrates naturally with trade finance, SME banking, FX hedging products, and supply-chain fintech, creating high-value data and upsell hooks.

Monetization Path

SaaS subscription for SMEs + white-label licensing for banks and trade platforms + premium advisory tier with analyst-augmented reports.

Cross-Industry Scalability

LintasNiaga's core engine is ready to be adapted for other industries reliant on complex, volatile global imports.

Future Planning & Roadmap

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Phase 1

- PP Resin (HS 3902.10) only Port Klang destination hub
- FOB quotes, up to 5 per run
- GLM extraction + AI reasoning
- Supabase + snapshot ingestion

Phase 2

- Persistent DB run state
- Add CIF/EXW incoterm support
- Richer ingestion scheduling
- Advanced hedge modeling

Phase 3

- Direct fintech execution workflows
- Complete deployment tooling,
- Real-time FX hooks
- API for platform partner

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Security & Data Planning

Data Isolation

Quote PDFs and extracted data are user-scoped. No cross-tenant data access. Uploaded documents stored in with per-user ACL.

Observability & Traceability

End-to-end auditability framework backed by structured logs, live-refreshed health monitoring, and a comprehensive per-run trace that archives the exact data and AI reasoning used for each recommendation.

Data Persistence

Audit trail assigns unique IDs to every analysis run, archiving the exact market snapshots and supplier quotes used. This ensures total transparency by documenting the logic and numerical simulations behind every recommendation.

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Thank You

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